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Exercise 1

1. Select *
From employees;

id	first_name	last_name	department	salary	hire_date	city
1	John	Doe	IT	55 000	2018-06-15	New York
2	Jane	Smith	HR	48 000	2019-07-20	Chicago
3	Mike	Johnson	Finance	60 000	2017-09-30	Los Angels
4	Sarah	Brown	IT	53 000	2021-03-25	New York
5	David	White	Marketing	52 000	2016-04-10	San Francisco
6	Emily	Davis	IT	62 000	2015-02-14	Chicago
7	Robert	Wilson	Finance	59 000	2019-10-01	Houston
8	Jessica	Moore	HR	51 000	2018-05-22	Los Angels
9	Daniel	Clark	Marketing	53 000	2022-06-01	Chicago
10	Laura	Hall	IT	50 000	2020-08-10	San Francisco

2. Select distinct department
From employees;

Department
IT
HR
Finance
Marketing

3. Select first_name,
last_name,
salary
From employees
Order by salary DESC;

3.

first_name	last_name	salary
Emily	Davis	62 000
Mike	Johnson	60 000
Robert	Wilson	59 000
John	Doe	55 000
Sarah	Brown	53 000
Daniel	Clark	53 000
David	White	52 000
Jessica	Moore	51 000
Laura	Hall	50000
Jane	Smith	48000

4.
```
Select  first_name,
        last_name,
        salary
From    employees
Order   by  salary  DESC
Limit   5 ;
```

first_name	last_name	salary
Emily	Davis	62 000
Mike	Johnson	60000
Robert	Wilson	59000
John	Doe	55000
Sarah	Brown	53000

5.
```
Select  first_name,
        last_name,
        department
From    employees
Where   department = 'IT' ;
```

5.

First-name	last-name	department
John	Doe	IT
Sarah	Brown	IT
Emily	Davis	IT
Laura	Hall	IT

6.
```
Select  first-name,
        department,
        salary
From    employees
Where   department = 'Finance'  AND  salary > 58000;
```

First_name	department	salary
Mike	Finance	60000
Robert	Finance	59000

7.
```
Select  first-name,
        department
From    employees
Where   department = 'HR'  OR  department = 'Marketing';
```

~~First~~

First_name	department
Jane	HR
David	Marketing
Daniel	Marketing
Jessica	HR

8.
```
Select  first-name,
        department
From    employees
Where   department  NOT IN ('IT');
```

8.

first_name	department
Jane	HR
Mike	Finance
David	Marketing
Robert	Finance
Jessica	HR
Daniel	Marketing

9.
```
Select    first_name
          department
From      employees
Where     department IN ('HR', 'IT')  OR  department = 'Finance';
```

first_name	department
John	IT
Jane	HR
Mike	Finance
Sarah	IT
Emily	IT
Robert	Finance
Jessica	HR
Laura	IT

10.
```
Select    first_name,
          department,
          salary, city
From      employees
Where     department = 'IT'  AND  salary > 50000
AND       city = 'New York';
```

10.

first_name	department	salary	city
John	IT	55000	New York
Sarah	IT	53000	New York

11. Select first_name,
last_name,
department,
salary
From employees
Where department = 'Finance' OR department = 'Marketing' AND
salary > 52000
Order by salary DESC;

first_name	last_name	department	salary
Mike	Johnson	Finance	60000
Robert	Wilson	Finance ~~Marketing~~	59000
Daniel	Clark	Marketing	53000

12. Select distinct city
From employees
Where department NOT IN ('IT', 'HR');

department	city
HR	Chicago
	Los Angels
	San Francisco

city
Chicago
Los Angels
San Francisco
Houston

13. Select first_name,
department,
salary,
hire date
From employees
Where department NOT IN ('Finance') AND salary > 50000
Order by hire_date ASC;

first_name	department	salary	hire_date
Emily	IT	62000	2015-02-14 ~~2016-04-10~~
David	Marketing	52000	2016-04-10
Jessica	HR	51000	2018-05-22
John	IT	55000	2018-06-15
Sarah	IT	53000	2021-03-25
Daniel	Marketing	53000	2022-06-01

14. Select first_name,
department,
city
From employees
Where city = 'Chicago' OR city = 'Los Angels AND department = 'IT'
OR department = 'Marketing'
Limit 3;

first_name	department	city
Emily	IT	Chicago
Daniel	Marketing	Chicago